

	plates.
4(a)(i)	Magnetic flux density is defined as the force acting on a conductor of <u>unit length</u>, carrying <u>unit current</u>, placed <u>perpendicularly</u> to the magnetic field. $B = \frac{F}{IL \sin 90^\circ} = \frac{F}{IL}$
4(a)(ii)	The magnetic flux linkage is the product of the number of turns of the coil of wire and the magnetic flux through each turn of the coil of wire, where the magnetic flux is the product of the component of the magnetic flux density normal to the plane of one turn of the coil and the area of this plane.
4(b)	At the instant when switch is closed, current in coil increases from zero to maximum. Magnetic flux linkage in the coil and <u>ring</u> increases. By Faraday's law of electromagnetic induction, the rate of change of magnetic flux linkage is the induced e.m.f. in the ring. As the ring is <u>closed</u>, an <u>induced current</u> flows. By Lenz's law, the induced current caused an opposing force to push the ring upwards.
4(c)	At instant of closing switch, an e.m.f. is induced in the ring.

	But <u>no induced current</u> flows in the insulator. Hence no opposing force can push the ring upwards.
Ex	<i>The root mean square (r.m.s.) value of an alternating current is the steady direct</i>